CONSECUTIVE MAPS IN A DISTINGUISHED TRIANGLE COMPOSE TO ZERO

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 582-600, following the statement at 574-580.

Editorial note. No omissions: the TR3 diagram is displayed; the remaining two identities are the TR2 rotation of the same argument.

Allowed standard prerequisites (editorial).

- Axioms of a pre-triangulated category. TR1, TR2, TR3 axioms of the chapter (stated in the section). Supplies the comparison diagram.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let (X, Y, Z, f, g, h) be a distinguished triangle. Then $g \circ f = 0$, $h \circ g = 0$ and $f[1] \circ h = 0$.*

Proof. By TR1 we know $(X, X, 0, 1, 0, 0)$ is a distinguished triangle. Apply TR3 to

$$\begin{array}{ccccccc} X & \longrightarrow & X & \longrightarrow & 0 & \longrightarrow & X[1] \\ \downarrow{\scriptstyle 1} & & \downarrow{\scriptstyle f} & & \vdots & & \downarrow{\scriptstyle 1[1]} \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \xrightarrow{h} & X[1] \end{array}$$

Of course the dotted arrow is the zero map. Hence the commutativity of the diagram implies that $g \circ f = 0$. For the other cases rotate the triangle, i.e., apply TR2. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.